Stroke

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In-Hospital Stroke Quality and Outcomes in the Mechanical Thrombectomy Era: Get With The Guidelines-Stroke, 2016 to 2023

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BACKGROUND: In-hospital stroke is associated with delayed recognition and worse outcomes compared with community-onset stroke. Contemporary national data in the mechanical thrombectomy (MT) era are limited.

METHODS: We performed a retrospective cohort study of adult ischemic stroke admissions in the Get With The Guidelines-Stroke registry from January 2016 to December 2023 (the MT era). In-hospital stroke was compared with community-onset stroke as the primary analysis. A secondary descriptive analysis comparing in-hospital patients with stroke from January 2006 to April 2012 and January 2016 to December 2023 was also performed. Primary outcomes were in-hospital mortality, discharge home, and independent ambulation at discharge. Multivariable logistic regression with generalized estimating equations accounted for within-hospital clustering and adjusted for demographics, vascular risk factors/comorbidities, and hospital characteristics. Temporal trends in MT use were assessed with the Cochran-Armitage trend test.

RESULTS: Among 4 996 392 ischemic stroke admissions from 2016 to 2023, 191 355 (3.8%) were in-hospital, and 4 805 037 (96.2%) were community onset. In-hospital patients presented with greater severity (National Institutes of Health Stroke Scale score >20: 14.5% versus 7.9%; National Institutes of Health Stroke Scale score, 0–4: 43.3% versus 60.4%) had worse outcomes including higher in-hospital mortality (adjusted odds ratio, 2.27 [95% CI, 2.18–2.36]; $P<0.001$), lower likelihood of discharge home (adjusted odds ratio, 0.46 [95% CI, 0.45–0.48]; $P<0.001$), and lower independent ambulation at discharge (adjusted odds ratio, 0.52 [95% CI, 0.50–0.53]; $P<0.001$). Recognition-to-computed tomography time was longer for in-hospital (median 51 [interquartile range, 16–269] versus 18 [10–39] minutes; $P<0.001$). MT rates increased significantly from 2016 to 2023 in both in-hospital and community-onset patients (4.47%–9.54% and 2.35%–5.55%, respectively; $P<0.001$). Quality metrics and outcomes significantly improved in the MT era, independent of treatment modality.

CONCLUSIONS: In patients with in-hospital stroke, MT rates increased over time, and improvements in quality metrics were observed regardless of treatment modality. A higher level of stroke severity and worse outcomes persists.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: benchmarking ■ goals ■ ischemic stroke ■ quality of health care ■ thrombectomy

In-hospital stroke (IHS) describes acute stroke symptoms first recognized after hospital arrival or among patients who are already in a health care setting.[1] Each year, the United States witnesses ≈35 000 to 75 000 IHS cases, accounting for 2% to 17% of all strokes.[1–4] Most IHSs are ischemic in nature (ranging from 89% to 98%), while the remainder are hemorrhagic.[5,6] IHS is associated with greater morbidity and mortality compared with community-onset stroke (COS).[7]

Moreover, evaluation and treatment of IHS are often challenged by atypical presentations, delays in recognition, variability in resource availability, and limited access

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Nonstandard Abbreviations and Acronyms

AHA	American Heart Association
COS	community-onset stroke
GWTG-Stroke	Get With The Guidelines-Stroke
IHS	in-hospital stroke
LDL	low-density lipoprotein
MT	mechanical thrombectomy
NIHSS	National Institutes of Health Stroke Scale
tPA	tissue-type plasminogen activator

to time-sensitive interventions. Despite these high stakes, IHS remains comparatively understudied. A recent American Heart Association (AHA) scientific statement underscored the lack of validated risk-stratification tools and large-scale, multivariable studies focused on IHS while outlining best practices aimed at improving care in this population.[7]

Over the past decade, national initiatives such as the AHAs Target: Stroke program have significantly improved acute stroke care for patients with COS, including reductions in symptom onset-to-needle times, improved thrombolysis rates, and better outcomes.[8,9] However, these efforts were not designed for, and have not been systematically evaluated in, patients with IHS. The extent to which such system-wide improvements have translated to patients experiencing stroke while already in the hospital remains uncertain.

In 2014, Cumbler et al[1] were the first to report disparities in quality of care and outcomes between IHS and COS using a national data set. The analysis was limited to ischemic stroke and was conducted in the pre-mechanical thrombectomy (MT) era, from January 2006 to April 2012. In 2015, 5 landmark randomized controlled trials, MR CLEAN (Multicenter Randomized Clinical Trial of Endovascular Treatment for Acute Ischemic Stroke in the Netherlands), ESCAPE (Endovascular Treatment for Small Core and Proximal Occlusion Ischemic Stroke), SWIFT PRIME (Solitaire With the Intention for Thrombectomy as Primary Endovascular Treatment), EXTEND-IA (Extending the Time for Thrombolysis in Emergency Neurological Deficits – Intra-Arterial), and REVASCAT (Randomized Trial of Revascularization With Solitaire FR Device Versus Best Medical Therapy in the Treatment of Acute Stroke due to Anterior Circulation Large Vessel Occlusion Presenting Within Eight Hours of Symptom Onset), established MT as a safe and effective treatment for acute ischemic stroke due to large-vessel occlusion within 6 hours of onset.[10–14] Subsequently, the DAWN (Diffusion-Weighted Imaging or Computed Tomography Perfusion Assessment With Clinical Mismatch in the Triage of Wake-Up and Late Presenting Strokes Undergoing Neurointervention With Trevo)[15] and DEFUSE 3 (Endovascular Therapy Following Imaging Evaluation for Ischemic Stroke 3)[16] trials expanded the MT treatment window up to 24 hours in select patients. The 2019 update to the AHA guidelines endorsed MT as the standard of care for eligible patients, mandating treatment within 6 hours, recommending it up to 16 hours, and permitting it up to 24 hours.[17] These changes have significantly altered the landscape of ischemic stroke treatment and outcomes compared with the pre-MT era.

Using contemporary data from the AHA Get With The Guidelines-Stroke (GWTG-Stroke) registry, we evaluated the presentation, quality metrics, treatment use, and outcomes of IHS compared with COS during the MT era, defined in our study as January 2016 to December 2023. Historical IHS data from the pre-MT era are presented only as contextual benchmarking and not as the primary focus of the current analysis. Our goal was to define the contemporary IHS care gap in the MT era and identify ongoing opportunities to improve timely recognition and treatment in hospitalized patients.

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METHODS

The data that support the findings of this study are available from the AHA Get With The Guidelines-Stroke registry under a data use agreement and are not publicly available. Requests from qualified researchers trained in human subject confidentiality protocols may be directed to qualityresearch@heart.org.

Study Design and Data Source

We conducted a retrospective cohort study using data from the GWTG-Stroke registry, which includes prospective, standardized clinical data submitted by participating US hospitals. The study period spanned January 1, 2016, through December 31, 2023. For contextual benchmarking, selected IHS measures from January 2006 to April 2012 were compared with the 2016 to 2023 cohort and were not included in the primary comparative or regression analyses. GWTG-Stroke is a national quality-improvement initiative designed to monitor and enhance adherence to evidence-based guidelines in stroke care. Data are abstracted by trained personnel using standardized case definitions and protocols. Participating hospitals obtained institutional review board approval or a waiver of informed consent for the collection of deidentified registry data. This analysis was conducted under a data use agreement and approved by the institutional review board. The study adhered to the STROBE reporting guidelines (Strengthening the Reporting of Observational Studies in Epidemiology; Supplemental Material).[18]

Study Population

The analysis included adult patients aged 18 to >90 years with a final diagnosis of ischemic stroke recorded in GWTG-Stroke during the study period. Patients transferred from another

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acute-care hospital were included only at the final destination hospital, consistent with registry conventions, to avoid double-counting across facilities. However, the registry extract used for this analysis did not permit consistent quantification of transfer-in admissions across sites and years; therefore, we were unable to report transfer-in frequency or conduct transfer-restricted sensitivity analyses, and this limitation is acknowledged. Consistent with prior work, we excluded sites that contributed no IHS cases during the study period (17 027 [0.34%] patients from 253 sites), yielding 4 996 392 patients from 2542 sites.

Classification of Stroke Onset

Stroke onset context was classified as IHS or COS based on the timing of symptom recognition relative to hospital arrival. IHS was defined as cases in which stroke symptoms were first recognized after hospital arrival (postarrival symptom recognition), whereas COS was defined as symptoms recognized before arrival. COS included patients with symptoms recognized before arrival, even if diagnostic confirmation occurred later during the hospitalization. For IHS, onset time was operationalized as the time of symptom recognition documented by hospital staff because symptom onset is frequently unknown in hospitalized patients. Stroke severity was assessed using the National Institutes of Health Stroke Scale (NIHSS) score when available and was incorporated in prespecified sensitivity models.

Outcomes

Primary outcomes were in-hospital mortality, discharge to home, and independent ambulation at discharge. Among patients with ischemic stroke treated with intravenous thrombolysis or MT, additional safety outcomes included symptomatic intracerebral hemorrhage within 36 hours, serious systemic hemorrhage within 36 hours, and other serious complications as defined in the GWTG-Stroke registry data dictionary. All outcomes were measured during the index hospitalization using standardized registry fields.

Quality metrics were based on established GWTG-Stroke performance measures.[19] Eligibility-based achievement measures included receipt of IV thrombolysis (IV tPA [tissue-type plasminogen activator]), early antithrombotic therapy within 48 hours, deep vein thrombosis prophylaxis within 48 hours in nonambulatory patients, antithrombotic therapy at discharge, anticoagulation at discharge for atrial fibrillation/flutter (when eligible), lipid-lowering therapy at discharge for patients with elevated LDL (low-density lipoprotein) cholesterol or prior statin use, and smoking cessation counseling for current or recent smokers. Process-of-care measures included recognition-to-computed tomography for IHS and arrival-to-computed tomography for COS, as well as recognition-to-needle ≤60 minutes for IHS and door-to-needle ≤60 minutes for COS among intravenous thrombolysis–treated patients with complete and valid timestamp data. Because arrival-based thrombolysis metrics are not directly applicable to IHS, we report overall intravenous thrombolysis receipt as the primary thrombolysis achievement measure and present time-based thrombolysis process measures separately when required time elements were available. We calculated a composite achievement measure (opportunity composite) defined as the total number of achievement measure interventions performed among eligible patients divided by the total number of possible achievement measure interventions among eligible patients. We also defined a defect-free (all-or-none) AHA-GWTG-Stroke summary measure reflecting the proportion of patients who received all of the achievement measures for which they were eligible.[19] For IHS, time zero was documented symptom recognition or stroke alert activation; for COS, time zero was hospital arrival. Two summary measures were evaluated: a composite achievement score and defect-free care. Eligible denominators were measure-specific and depended on inclusion/exclusion criteria, availability of required data elements, and documented contraindications or exclusions; accordingly, denominators varied across measures, between onset groups, and across registry epochs. Detailed definitions, numerators, denominators, and exclusions for each measure are provided in the Supplemental Material.

Statistical Analysis

Categorical variables were summarized using frequencies and percentages, and continuous variables were reported as medians with interquartile ranges. Given the large sample size and multiple comparisons across numerous baseline and process measures, group differences are presented descriptively, and standardized differences were used to quantify baseline covariate imbalance. Multivariable logistic regression models with generalized estimating equations were used to estimate adjusted odds ratios with 95% CIs, accounting for clustering at the hospital level using an exchangeable correlation structure. Prespecified covariates included age, sex, race and ethnicity, vascular risk factors and comorbidities (eg, hypertension, diabetes, dyslipidemia, atrial fibrillation/flutter, coronary artery disease/prior myocardial infarction, heart failure, prior stroke, or transient ischemic attack), smoking status, and hospital characteristics (geographic region, academic status, annual stroke volume, and bed count). NIHSS score was included in sensitivity analyses among patients with nonmissing data. Temporal trends in MT utilization and other variables from 2016 to 2023 were assessed using the Cochran-Armitage trend test. Statistical analyses were performed using SAS, version 9.4 (SAS Institute, Inc). Prespecified secondary and sensitivity analyses included NIHSS score–adjusted analyses, exclusion of elective carotid procedures, alternate definitions of symptomatic intracerebral hemorrhage where applicable, complete-case analyses, and subgroup or interaction analyses by age, sex, race and ethnicity, hospital type, care setting, and geographic region.

RESULTS

A total of 4 996 392 admissions with a final diagnosis of ischemic stroke were identified from 2542 participating US hospitals in GWTG-Stroke between January 2016 and December 2023, after excluding sites that contributed no IHS cases. Of these, 191 355 (3.8%) were classified as IHS and 4 805 037 (96.2%) as COS. The flow of cohort assembly is presented in Figure 1.

Demographic and baseline clinical characteristics of patients in each cohort are summarized in Table 1. Age distributions were comparable between groups (patients aged <50 years: 9.7% in both groups; P<0.001; 50–89 years: 84.4% in IHS versus 83.6% in COS; P<0.001; and

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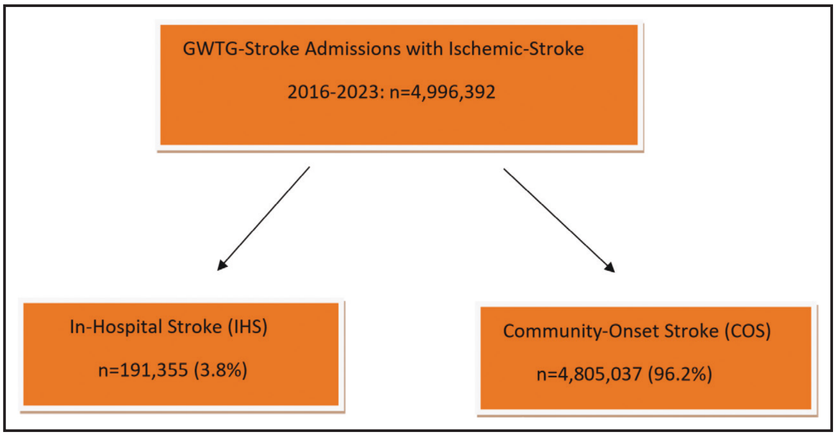

Figure 1. Cohort assembly for ischemic stroke admissions in Get With The Guidelines (GWTG)-Stroke, 2016 to 2023.
Among 4 996 392 ischemic stroke admissions in the analytic cohort, 191 355 (3.8%) were classified as in-hospital stroke (IHS) and 4 805 037 (96.2%) as community-onset stroke (COS). Sites contributing no IHS cases during the study period were excluded from the final analytic cohort.

≥90 years: 5.9% versus 6.7%; P<0.001), and the proportion of women was slightly higher in IHS (50.4% versus 49.7%; P<0.001). Patients with IHS had a greater burden of thromboembolic and cardiovascular comorbidities than those with COS, including atrial fibrillation/flutter (20.7% versus 16.6%; P<0.001), coronary artery disease/prior myocardial infarction (26.6% versus 20.2%; P<0.001), and heart failure (14.3% versus 9.0%; P<0.001). Carotid stenosis (3.90% versus 3.07%; P<0.001) and prosthetic heart valves (1.77% versus 1.21%; P<0.001) were also more common among patients with IHS. Ischemic stroke etiologies differed between groups. Cardioembolic stroke was more frequently assigned in IHS (33.1% versus 23.7%; P<0.001), whereas small-vessel occlusion (13.9% versus 21.4%; P<0.001) and cryptogenic stroke classification (25.9% versus 30.4%; P<0.001) were less common in IHS. Preadmission medication patterns differed between groups: patients with IHS were more frequently taking antiplatelet and anticoagulant therapy, whereas antihypertensive, cholesterol-reducing, and antihyperglycemic medication use was modestly lower compared with COS. Functional status before admission was lower in IHS, with fewer patients ambulating independently (85.3% versus 91.4%; P<0.001) and more requiring assistance (8.9% versus 5.6%; P<0.001) or being nonambulatory (5.8% versus 3.0%; P<0.001).

IHS was associated with greater neurological severity at presentation. NIHSS scores >20 were observed more frequently in IHS than COS (14.5% versus 7.9%; P<0.001), while NIHSS scores 0 to 4 were less common (43.3% versus 60.4%; P<0.001). Discharge disability was worse among patients with IHS, with more patients discharged with severe disability (modified Rankin Scale score, 5: 19.8% versus 11.8%; P<0.001) and fewer discharged without symptoms (modified Rankin Scale score, 0: 11.4% versus 17.8%; P<0.001). Full severity and disability distributions are detailed in Tables S1 and S2.

When assessed using GWTG-Stroke process-of-care measures, patients with IHS showed differences across several domains (Table 2). Time to initial brain imaging was longer in IHS than COS (median, 51 [interquartile range, 16–269] versus 18 [interquartile range, 10–39] minutes; P<0.001). Early antiplatelet administration was lower among patients with IHS (85.2% versus 92.9%; P<0.001), whereas early anticoagulation use was more frequent (35.6% versus 27.0%; P<0.001). In the contemporary cohort, IV tPA was administered to 12 365 of 191 355 patients with IHS (6.46%) compared with 577 635 of 4 805 037 patients with COS (12.02%; P<0.001). Among patients treated with IV tPA, onset/recognition-to-needle ≤60 minutes occurred at similar rates in IHS and COS (82.2% versus 84.5%; P<0.001). Overall adherence to guideline-based care remained high in both groups, with opportunity-based composite performance exceeding 95% and defect-free care occurring in 88.4% of patients with IHS and 86.0% of patients with COS (P<0.001; Table 2). Figure 2 illustrates that catheter-based stroke treatment rates increased steadily over time in both groups, with consistently higher rates in IHS than COS across all study years; this upward linear trend was confirmed by the Cochran-Armitage trend test for both groups (P<0.001; Table S3). Multivariable logistic regression showed that each 1-year increase in admission year was associated with an 11% increase in the odds of receiving catheter-based stroke treatment (odds ratio, 1.11 [95% CI, 1.11–1.12]; P<0.001), while IHS had higher odds of receiving catheter-based stroke treatment than COS (odds ratio, 1.80 [95% CI, 1.77–1.83]; P<0.001; Table S4).

Regarding primary clinical outcomes, IHS was associated with a worse prognosis. In adjusted models, patients with IHS had higher in-hospital mortality (adjusted odds ratio, 2.27 [95% CI, 2.18–2.36]; P<0.001) and lower odds of discharge to home (adjusted odds ratio, 0.46 [95% CI, 0.45–0.48]; P<0.001) and independent ambulation at discharge (adjusted odds ratio, 0.52 [95% CI, 0.50–0.53]; P<0.001). These associations remained significant in models additionally adjusted for NIHSS score among patients with available severity data. Primary clinical outcomes, including crude rates and unadjusted/adjusted odds ratios with NIHSS score sensitivity

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Table 1. Comparison of Patient Characteristics Between In-Hospital Stroke and Community-Onset Stroke (2016–2023)

Variable	Category/reference	In-hospital stroke (N=191 355)	Community-onset stroke (N=4 805 037)	*P* value
Sex, n (%)	Female	96 327 (50.4)	2 389 062 (49.7)	<0.001
	Male	94 952 (49.6)	2 413 757 (50.3)	<0.001
Age group, y; n (%)	<50	18 517 (9.7)	466 502 (9.7)	0.66
	50–89	161 547 (84.4)	4 016 944 (83.6)	<0.001
	≥90	11 291 (5.9)	321 591 (6.7)	<0.001
Race, n (%)	Black	32 449 (17.0)	817 475 (17.0)	0.5644
	American Indian	1142 (0.6)	22 952 (0.5)	<0.001
	Asian	6870 (3.6)	163 172 (3.4)	<0.001
	Hispanic	16 412 (8.6)	407 184 (8.5)	0.1105
	Native Hawaiian/Pacific Islander	629 (0.33)	14 528 (0.30)	0.04
	White	135 622 (70.9)	3 465 855 (72.2)	<0.001
	Other/unknown	14 664 (7.7)	321 538 (6.7)	<0.001
Stroke cause, n (%)	Large-artery atherosclerosis	20 477 (19.77)	461 549 (19.7)	0.69
	Cardioembolism	34 303 (33.11)	554 833 (23.68)	<0.001
	Small-vessel occlusion	14 441 (13.94)	500 142 (21.37)	<0.001
	Dissection	893 (0.92)	18 266 (0.81)	<0.001
	Hypercoagulability	1791 (1.83)	21 223 (0.94)	<0.001
	Other determined cause	4403 (4.38)	65 845 (2.87)	<0.001
	Cryptogenic stroke	26 822 (25.89)	712 209 (30.42)	<0.001
	Undocumented	40 000 (27.86)	883 640 (27.40)	<0.001
Medical history, n (%)	Atrial fibrillation/flutter	39 305 (20.71)	789 570 (16.60)	<0.001
	Diabetes	69 419 (36.58)	1 570 661 (33.04)	<0.001
	E-cigarette use	169 (0.09)	4613 (0.10)	0.27
	Heart failure	27 204 (14.33)	427 220 (8.98)	<0.001
	Hypertension	142 456 (75.06)	3 521 000 (74.07)	<0.001
	CAD/prior MI	50 439 (26.58)	960 752 (20.21)	<0.001
	DVT/PE	7392 (3.89)	136 538 (2.87)	<0.001
	Drugs/alcohol abuse	15 843 (8.35)	371 561 (7.81)	<0.001
	Familial hypercholesterolemia	857 (0.45)	23 614 (0.50)	<0.001
	Migraine	4912 (2.59)	153 989 (3.24)	<0.001
	Previous TIA	11 586 (6.10)	388 535 (8.18)	<0.001
	Chronic renal insufficiency	26 837 (14.14)	462 131 (9.71)	<0.001
	Smoker	29 018 (15.29)	825 071 (17.36)	<0.001
	Carotid stenosis	7407 (3.90)	146 098 (3.07)	<0.001
	Dementia	4773 (2.51)	124 116 (2.61)	<0.001
	Depression	26 078 (13.74)	631 120 (13.28)	<0.001
	Dyslipidemia	87 566 (46.14)	2 218 200 (46.67)	<0.001
	Family history of stroke	9336 (4.92)	306 551 (6.44)	<0.001
	History of emerging infectious diseases	4088 (2.15)	66 112 (1.39)	<0.001
	Obesity/overweight	50 360 (26.54)	1 337 434 (28.14)	<0.001
	Prosthetic heart valve	3360 (1.77)	57 392 (1.21)	<0.001
	Sickle cell	211 (0.11)	3591 (0.08)	<0.001
	Previous stroke	40 429 (21.30)	1 103 957 (23.23)	<0.001
	Peripheral vascular disease	9745 (5.13)	169 341 (3.56)	<0.001
Ambulatory status, n (%)	Independent ambulation	111 331 (85.29)	3 037 101 (91.41)	<0.001
	Ambulation with assistance	11 627 (8.91)	185 124 (5.58)	<0.001
	Unable to ambulate	7577 (5.80)	99 886 (3.01)	<0.001

(*Continued*)

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Table 1. Continued

Variable	Category/reference	In-hospital stroke (N=191 355)	Community-onset stroke (N=4 805 037)	*P* value
Prestroke modified Rankin Scale score, n (%)	≤2	41 386 (82.07)	1 100 758 (86.98)	<0.001
	3–5	9040 (17.93)	164 783 (13.02)	<0.001
Medication before admission, n (%)	No medications	29 254 (15.29)	792 648 (16.50)	<0.001
	Antiplatelet	78 168 (41.56) Missing n=3254	1 852 314 (39.53) Missing n=120 097	<0.001
	Anticoagulant	31 911 (16.96) Missing n=3254	606 558 (12.94) Missing n=120 097	<0.001
	Antihypertensive	56 203 (33.72) Missing n=24 663	1 498 596 (37.22) Missing n=778 020	<0.001
	Cholesterol-reducer	99 966 (52.54) Missing n=1092	2 619 677 (54.87) Missing n=29 556	<0.001
	Antihyperglycemic	118 976 (71.16) Missing n=24 161	2 975 661 (73.72) Missing n=768 114	<0.001
	Antidepressant	124 182 (82.39) Missing n=40 625	3 007 891 (83.38) Missing n=1 197 696	<0.001

CAD indicates coronary artery disease; DVT, deep vein thrombosis; HF, heart failure; MI, myocardial infarction; PE, pulmonary embolism; and TIA, transient ischemic attack.

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analyses, are presented in Table S5. Thrombolysis-related safety outcomes among IV tPA–treated patients are presented in Table S6.

Finally, compared with historical data from earlier national analyses, IHS process performance improved substantially in the contemporary era, including defect-free care and timely thrombolysis metrics (Table 3). Era-based benchmarking of IHS clinical outcomes between 2016 to 2023 and 2006 to 2012, including independent ambulation at discharge, discharge to home, and in-hospital mortality, is summarized in Table S7.

DISCUSSION

The findings in our analysis highlight both progress in guideline-driven efforts to improve IHS care and the need to further reduce IHS-to-COS treatment variance. In this contemporary national GWTG-Stroke sample from January 2016 to December 2023, IHS accounted for ≈3.8% of ischemic stroke admissions. Compared with COS, IHS was associated with greater neurological severity, higher burden of cardiovascular comorbidities, and worse functional status before admission. These differences likely reflect the acute illness and physiological stressors prompting hospitalization and may not be fully modifiable through stroke-specific process improvements alone. After adjusting for demographics, comorbidities, and NIHSS scores, IHS remained independently associated with higher in-hospital mortality and lower odds of discharge to home and independent ambulation. These findings are consistent with prior prethrombectomy era observations and underscore that the outcome gap likely reflects a combination of patient factors and care delivery challenges unique to the inpatient setting.

Compared with the 2006 to 2012 period described by Cumbler et al,[1] process-of-care performance for patients with IHS has improved markedly in the contemporary era. As detailed in Table 3, key quality metrics have shown substantial gains. Whereas Cumbler et al[1] reported wide disparities in defect-free care (60.8% for IHS versus 82.0% for COS), our data demonstrate a notable convergence, with patients with IHS achieving 88.4% compared with 86.0% in COS. Similarly, the opportunity-based composite performance exceeded 95% in both groups, reflecting high overall adherence to guideline-recommended care. Eligible denominators are measure-specific and depend on the availability of required time elements and documentation of contraindications; therefore, measure-level comparisons, particularly for intravenous thrombolysis and deep vein thrombosis prophylaxis, should be interpreted in the context of differential measure eligibility and data completeness. These gains are consistent with the AHA scientific statement's emphasis on hospital-wide stroke training, standardized evaluation/imaging workflows, and IHS-specific quality oversight programs to improve timely recognition and treatment.[7] Despite these advances, gaps remain in specific measures: early antiplatelet administration and discharge antithrombotic therapy continue to lag in the IHS population. In addition, recognition-to-computed tomography times remained significantly prolonged for patients with IHS although onset-to-needle times ≤60 minutes were achieved at rates comparable to COS. These findings are consistent with registry trends in the postthrombectomy era, which show persistent but narrowing recognition-to-imaging delays among hospitalized patients, likely reflecting growing integration of endovascular stroke protocols.[4–7] Era-based benchmarking of IHS clinical outcomes (2016–2023 versus 2006–2012), including independent ambulation at discharge, discharge to home, and in-hospital mortality, is summarized in Table S7. Notably, much of the 2016 to

Table 2. GWTG-Stroke Achievement and Quality Measures

Variables	In-hospital strokes (N=191 355)	Community-onset strokes (N=4 805 037)	*P* value
Achievement measures			
Received IV tPA n (%), total n	12 365 (6.46) 191 355	577 635 (12.02) 4 805 037	<0.001
Early antiplatelet n (%), n eligible	90 819 (85.23) 106 562	2 517 732 (92.94) 2 708 915	<0.001
Early anticoagulant n (%), n eligible	37 958 (35.62) 106 562	730 863 (26.98) 2 708 915	<0.001
DVT prophylaxis n (%), n eligible	1322 (83.78) 1578	37 632 (89.39) 42 096	<0.001
Antithrombotics on discharge n (%), n eligible	123 885 (71.55) 173 154	3 385 027 (81.27) 4 165 096	<0.001
Anticoagulants on discharge for atrial fibrillation/flutter n (%), n eligible	25 629 (54.34) 47 161	535 752 (59.19) 905 192	<0.001
Statin for LDL >100 (or if LDL not determined) n (%), n eligible	32 844 (77.13) 42 580	1 253 751 (83.22) 1 506 556	<0.001
Smoking cessation counseling n (%), n eligible	8943 (97.19) 9202	619 999 (97.64) 634 930	<0.001
Quality measures			
Recognition-to-CT time* in patients presenting with stroke symptoms	Median (IQR)=51 (16–269) n eligible=22 575	Median (IQR)=18 (10–39) n eligible=2 003 793	<0.001
LDL 100 or ND n (%), n eligible	29 416 (98.06) 29 998	2 260 381 (98.37) 2 297 771	<0.001
Dysphagia screening n (%), n eligible	49 263 (85.26) 57 778	2 997 475 (84.41) 3 551 105	<0.001
Stroke education n (%), n eligible	121 588 (92.70) 131 165	3 457 121 (95.35) 3 625 878	<0.001
Rehabilitation assessment n (%), n eligible	132 294 (98.15) 134 785	3 649 597 (97.35) 3 748 769	<0.001
Recognition/alert-to-IV tPA* in 60 min n (%), n eligible†	643 (82.23) 782	269 784 (84.61) 318 840	<0.001
LDL documented n (%), n eligible	39 023 (92.08) 42 379	2 886 586 (93.53) 3 085 512	<0.001
Intensive statin therapy n (%), n eligible	33 698 (86.61) 38 908	2 476 283 (86.99) 2 846 609	0.03
Composite achievement measure‡			<0.001
Achievement measures performed/achievement measure opportunities	Mean=95.59% n eligible=67 578	Mean=95.00% n eligible=3 976 908	
Defect-free achievement measure§ n (%), n eligible	59 752 (88.42) 67 578	3 421 928 (86.04) 3 976 908	<0.001

COS indicates community-onset stroke; CT, computed tomography; DVT, deep vein thrombosis; GWTG, Get With The Guidelines; IHS, in-hospital stroke; IQR, interquartile range; LDL, low-density lipoprotein; ND, not determined; and tPA, tissue-type plasminogen activator.

*For IHS, time intervals (recognition-to-CT and recognition-to-IV tPA) are referenced to documented symptom recognition/stroke alert activation (time zero); for COS, they are referenced to hospital arrival.

†For this time-based measure, the eligible denominator includes only IV tPA–treated patients with IHS with complete and valid timestamp elements required to calculate recognition/alert-to-IV tPA time (ie, documented recognition/alert time and IV tPA initiation time).

‡Composite achievement measure is defined as the total number of achievement measure interventions performed among eligible patients divided by the total number of possible achievement measure interventions among eligible patients.

§Defect-free achievement measure includes all achievement measures and reflects the proportion of patients who received all of the achievement measures for which they were eligible.

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2023 period preceded publication and likely widespread implementation of the AHA scientific statement outlining IHS-specific best practices in February 2022.

Intravenous thrombolysis use was less frequent in IHS compared with COS, which may reflect differences in monitoring, eligibility, contraindication profiles, and documentation among hospitalized patients. Among treated patients, the safety profile of IV thrombolytic remained similar between groups. Rates of symptomatic intracerebral hemorrhage and serious bleeding did not differ significantly, aligning with both prethrombectomy GWTG-Stroke data and recent meta-analysis showing comparable risks across onset settings.[1,20] These findings support the safety of appropriately selected patients with IHS for thrombolytic therapy and reinforce the importance of timely recognition and intervention.

MT use increased markedly over the study period, consistent with national trends following guideline updates.[20] In our registry, MT use increased over time in both groups but remained consistently higher in IHS than in COS throughout the study period (Figure 2; Table S3). This finding is reinforced by prior observational studies showing greater endovascular therapy utilization in IHS populations, likely reflecting greater stroke severity and a higher prevalence of large-vessel occlusion among hospitalized patients. In a nationwide Chinese Stroke Center Alliance registry, MT was more frequent in IHS than in COS (4.2% versus 1.3%).[21] Similarly, Liu et al[22] reported higher MT use in IHS compared with COS in a tertiary-center cohort (10.4% versus 2.0%), despite lower intravenous thrombolysis rates in the inpatient population. Although our analysis was not designed to evaluate MT-specific outcomes directly, Lu et al[23] reported comparable rates of successful recanalization, symptomatic intracerebral hemorrhage, mortality, and 90-day functional independence between MT-treated patients with IHS and COS, suggesting that timely thrombectomy within an IHS-specific protocol may help mitigate some outcome differences. In contrast, the persistently worse outcomes observed in our broader IHS

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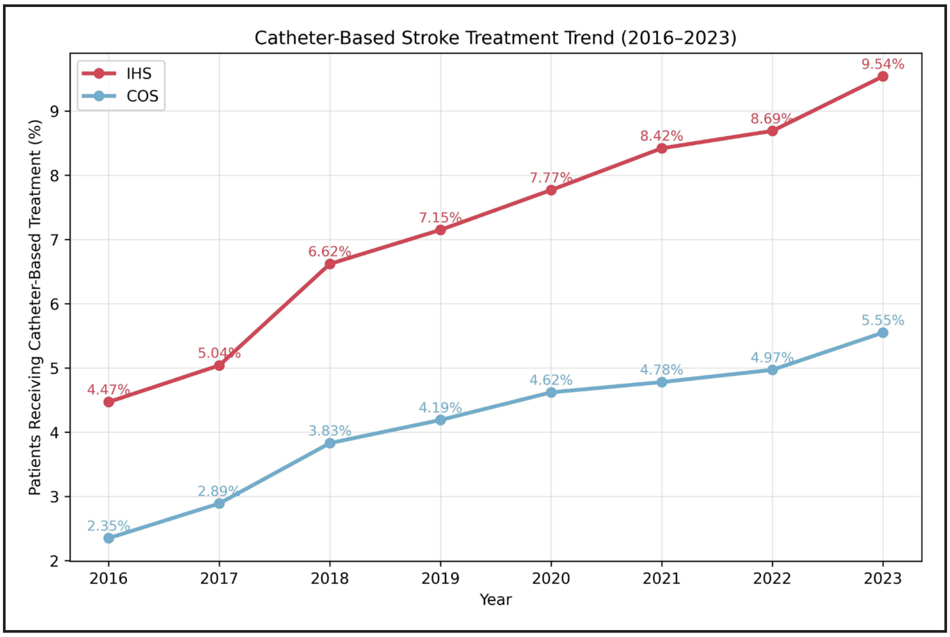

Figure 2. Temporal trends in catheter-based stroke treatment in in-hospital stroke (IHS) and community-onset stroke (COS), 2016 to 2023.
The Cochran-Armitage trend test P<0.001. Catheter-based stroke treatment rates increased steadily from 2016 to 2023 in both onset groups, with consistently higher rates in IHS throughout the study period. Treatment rates increased from 4.47% to 9.54% in IHS and from 2.35% to 5.55% in COS; Cochran-Armitage trend test P<0.001 for both groups.

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cohort likely reflect differences in baseline stroke severity, medical complexity, and early system-level delays. Importantly, system-level changes accompanying MT adoption may have contributed to broader improvements in acute stroke workflows; however, outcomes after IHS remain strongly influenced by underlying illness severity and inpatient care complexity. Despite improvements in guideline adherence, outcome disparities for IHS persist. Contributing factors likely include atypical or delayed symptom recognition due to sedation, intubation,

Table 3. Comparison of GWTG-Stroke Achievement and Quality Measures of In-Hospital Stroke Between Nouh et al Study (2016–2023) and Cumbler et al Study (2006–2012)

Variable	In-hospital strokes between 2016 and 2023 (Nouh et al study; N=191 355)	In-hospital strokes between 2006 and 2012 (Cumbler et al study; N=21 349)	*P* value
Achievement measures			
DVT prophylaxis n (%), n eligible	1322 (83.78) 1578	7810 (88.8%) 8793	<0.001
Antithrombotics on discharge n (%), n eligible	123 885 (71.55) 173 154	13 658 (96.1) 14 208	<0.001
Anticoagulants on discharge for atrial fibrillation/flutter n (%), n eligible	25 629 (54.34) 47 161	3089 (90.6) 3410	<0.001
Smoking cessation counseling n (%), n eligible	8943 (97.19) 9202	2010 (92.2) 2181	<0.001
Quality measures			
Dysphagia screening n (%), n eligible	49 263 (85.26) 57 778	10 692 (65.6) 16 302	<0.001
Rehabilitation assessment n (%), n eligible	132 294 (98.15) 134 785	14 442 (96.8) 14 992	<0.001
Recognition/alert-to-IV tPA* in 60 min n (%), n eligible†	643 (82.23) 782	421 (19.7) 2132	<0.001
Received IV tPA, n (%) total n	12 365 (6.46) 191 355	2352 (11.0) 21 349	<0.001
LDL documented n (%), n eligible	39 023 (92.08) 42 379	10 641 (71.5) 14 890	<0.001
Intensive statin therapy n (%), n eligible	33 698 (86.61) 38 908	673 (20.8) 3231	<0.001
Defect-free achievement measure n (%), n eligible	59 752 (88.42) 67 578	11 334 (60.8) 18 628	<0.001

COS indicates community-onset stroke; CT, computed tomography; DVT, deep vein thrombosis; GWTG, Get With The Guidelines; IHS, in-hospital stroke; LDL, low-density lipoprotein; and tPA, tissue-type plasminogen activator.

*For IHS, time intervals (recognition-to-CT and recognition-to-IV tPA) are referenced to documented symptom recognition/stroke alert activation (time zero); for COS, they are referenced to hospital arrival.

†For this time-based measure, the eligible denominator includes only IV tPA–treated patients with IHS with complete and valid timestamp elements required to calculate recognition/alert-to-IV tPA time (ie, documented recognition/alert time and IV tPA initiation time).

overlapping medical issues, concurrent systemic illness, and inflammatory or periprocedural physiological stress. System-level contributors may include logistical constraints in mobilizing imaging and response teams outside the emergency department and limited staff familiarity with stroke recognition protocols on nonneurology services. Moreover, patients with IHS may remain under the primary admitting service (eg, cardiology or surgery) with neurology consultation rather than transfer to a neurology service. In this context, periprocedural bleeding risk and competing inpatient priorities may influence discharge antithrombotic decision-making and the documentation of contraindications within registry fields. Data from the Ontario Stroke Registry echo these findings, with IHS associated with longer recognition-to-imaging and recognition-to-thrombolysis intervals, lower thrombolysis use, and worse discharge functional outcomes, even after adjustment for age, comorbidities, stroke type, and stroke severity.[5]

Our findings support the need for IHS-specific quality-improvement initiatives rather than relying solely on aggregated hospital-level stroke metrics, which may obscure inpatient care gaps.[24,25] Several strategies are feasible and aligned with current AHA recommendations: proactive surveillance and early recognition on high-risk units (eg, postoperative, cardiac, and intensive care unit) with empowered rapid response teams authorized to activate code stroke protocols; routine reporting of IHS-specific metrics, including recognition-to-computed tomography, recognition-to-needle, and defect-free care, stratified by onset location; targeted training and simulation for nonneurology clinical teams to reduce recognition-to-imaging delays; and thrombectomy pathway auditing with timestamped tracking from symptom recognition through angiography to identify and address early phase delays unique to the inpatient setting.[7] These recommendations are supported by systems-level studies linking organized acute stroke pathways to improved reperfusion performance and outcomes, and are increasingly viewed as essential for equitable care delivery across all patients with stroke.

Strengths of this analysis include its large contemporary multicenter cohort, standardized registry definitions, and modeling approaches that account for within-hospital clustering with sensitivity analyses incorporating NIHSS score. Limitations include the voluntary nature of registry participation, potential under-ascertainment or misclassification of IHS events, and incomplete capture of key time fields and NIHSS score for a subset of admissions. Time-based thrombolysis measures were available only for a subset with complete timestamp elements; therefore, eligible denominators differed between cohorts and eras, and comparisons of ≤60-minute metrics should be interpreted cautiously. The registry extract also did not permit consistent quantification of transfer-in admissions or site-level dispersion of IHS reperfusion therapy use across hospitals and years, limiting transfer-restricted analyses and granular hospital-level treatment distribution assessments. In addition, the analytic window ended on December 31, 2023, to maximize data completeness and stability; when this project was initiated in early 2025, registry data for 2024 were not yet sufficiently complete and remained subject to delayed abstraction/submission and subsequent data cleaning, which could bias year-to-year comparisons. Finally, long-term functional outcomes are not captured in this data set, and as an observational analysis, unmeasured confounding and treatment selection bias cannot be excluded.

CONCLUSIONS

In the MT era, IHS remains associated with greater neurological severity, delayed imaging, and worse in-hospital outcomes than COS. Although adherence to the GWTG-Stroke measures was high and MT use increased over time in both groups, meaningful disparities persist. Improvements in quality-based care metrics were, however, achieved in patients with IHS during the MT era, regardless of treatment modality. Future work should prioritize IHS-specific workflows, recognition-based time metrics, and targeted quality improvement while acknowledging that a substantial portion of outcome disparity reflects baseline illness severity, periprocedural risk, and limited eligibility for reperfusion therapy.

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Supplemental Material

Supplemental Methods
Tables S1–S7
STROBE Checklist

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